IT-Simplera Solutions

Week 4 Task Report

Enterprise Network Security and Access Control

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Enterprise Network Security & Access Control Lists

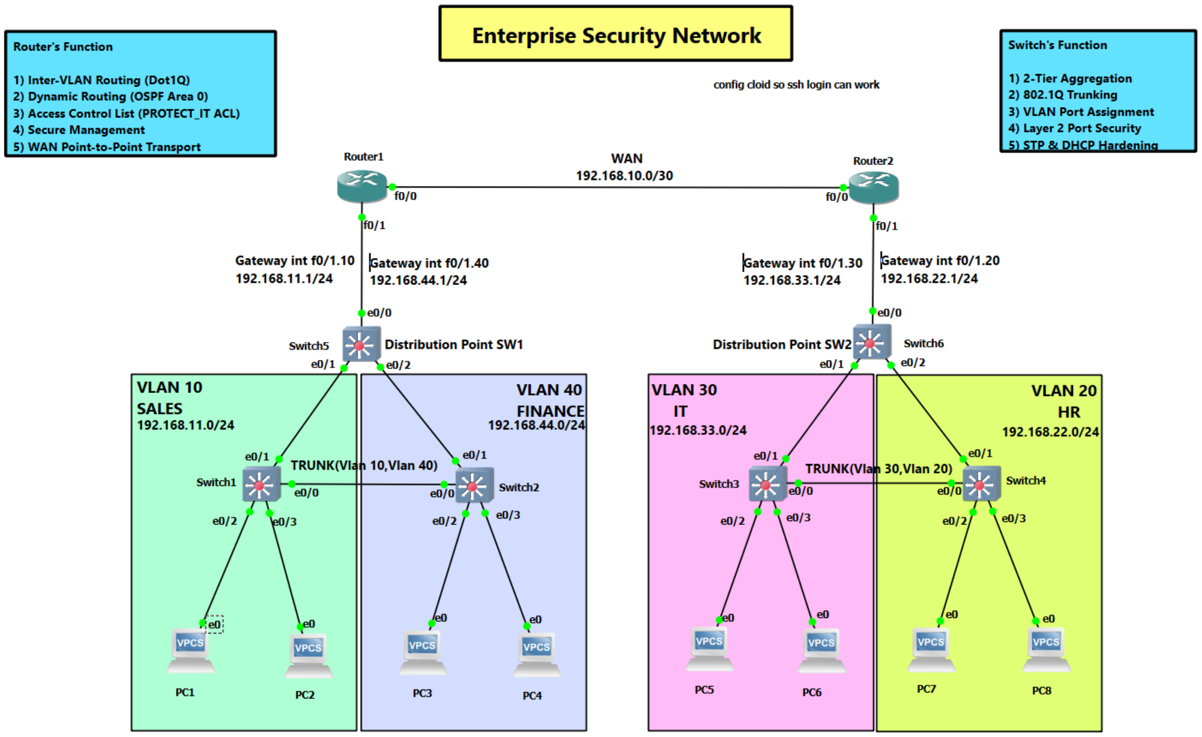
Comprehensive Technical Documentation, Verification & Troubleshooting Report

1. Network Topology & Device Addressing

The network topology consists of a multi-router core (R1 and R2), distribution/access switches (SW1, SW2, SW3, SW4, DSW1, DSW2), and logically isolated departmental VLANs. Inter-VLAN routing is configured via sub-interfaces (Router-on-a-Stick), and traffic flow between subnets is strictly enforced using Access Control Lists (ACLs).

Device	Interface / Sub-if	Department / VLAN	IP Address	Subnet Mask	Role / Description
R1	Fa0/1.10	Sales (VLAN 10)	192.168.11.1	255.255.255.0	Default Gateway for Sales
R1	Fa0/1.40	Finance (VLAN 40)	192.168.44.1	255.255.255.0	Default Gateway for Finance
R2	Fa0/1.30	IT (VLAN 30)	192.168.33.1	255.255.255.0	Default Gateway for IT
R2	Fa0/1.20	HR (VLAN 20)	192.168.22.1	255.255.255.0	Default Gateway for HR
R1 - R2	Fa0/0	WAN	192.168.10.0	255.255.255.0	Link WAN
DSW1/2 - R1/R2	Eth0/0	Gateway	192.168.11.1, 192.168.44.1, 192.168.22.2, 192.168.33.2	255.255.255.0	Link Trunk Gateway
DSW1/2 – SW1/2/3/4	Eth0/1, Eth0/2	Trunk	VLAN 10, 20, 30, 40		Trunks to VLAN's SW
SW1 – SW4, SW3 – SW2	Eth0/0	Trunk	VLAN 10 – VLAN 40, VLAN 20 - VLAN 30		Trunks BTW VLAN SW
SW4	Vlan20	HR (VLAN 20)	192.168.22.250	255.255.255.0	Management Switch IP
IT-PC	Eth0/2, Eth0/3	IT (VLAN 30)	192.168.33.2, 192.168.33.3	255.255.255.0	IT Department Host
HR-PC	Eth0/2, Eth0/3	HR (VLAN 20)	192.168.22.2, 192.168.22.3	255.255.255.0	HR Management Host

Sales-PC	Eth0/2, Eth0/3	Sales (VLAN 10)	192.168.11.2, 192.168.11.3	255.255.255.0	Sales Department Host
Finance-PC	Eth0/2, Eth0/3	Finance (VLAN 44)	192.168.44.2, 192.168.44.3	255.255.255.0	Finance Department Host



2. Core Protocols & Technical Concepts Explained

2.1 Inter-VLAN Routing (Router-on-a-Stick)

Inter-VLAN routing allows logically separated VLANs on Layer 2 switches to communicate at Layer 3 by utilizing virtual sub-interfaces on a single physical router port. Sub-interfaces Fa0/1.10,Fa0/1.20, Fa0/1.30, and Fa0/1.40 are configured with 802.1Q encapsulation to serve as default gateways for their respective VLANs.

2.2 Standard vs. Extended Named Access Control Lists (ACLs)

- **Standard ACLs (1-99, 1300-1999):** Standard ACLs filter traffic based exclusively on the Source IP address. They are traditionally placed as close to the destination as possible.
- **Extended ACLs (100-199, 2000-2699):** Extended ACLs filter traffic based on Source IP, Destination IP, Protocol type (IP, TCP, UDP, ICMP), and Port numbers. They are placed as close to the source as possible.
- **Named ACLs:** Named ACLs allow using descriptive string names instead of numbers for enhanced readability, management, and individual line editing.

2.3 Secure Shell (SSH v2)

SSH v2 encrypts all remote management terminal traffic across the network, completely replacing insecure plaintext Telnet and protecting administrative credentials from eavesdropping. Enforced across all VTY lines (0 4) using 1024-bit RSA key encryption.

2.4 Layer 2 Infrastructure Security Features

- **Port-Security:** Restricts switch port connectivity by authorizing specific MAC addresses and executing violation actions (Protect, Restrict, Shutdown) upon unauthorized connections.
- **DHCP Snooping:** Mitigates rogue DHCP server attacks by filtering untrusted DHCP messages and building a binding table of valid IP-to-MAC mappings.
- **Spanning-Tree PortFast:** Immediately transitions end-user access ports to the STP forwarding state, bypassing listening/learning phases.
- **STP BPDU Guard:** Disables a PortFast-enabled port instantly if an unexpected Bridge Protocol Data Unit (BPDU) is detected.
- **STP Root Guard:** Prevents unauthorized downstream switches from claiming the Root Bridge role in the STP topology.

3. Step-by-Step Security Configurations

3.1 Router Hardening & SSH Setup (R1 & R2)

```
R1(config)# ip domain-name enterprise.local
R1(config)# crypto key generate rsa general-keys modulus 1024
R1(config)# ip ssh version 2
R1(config)# username admin privilege 15 secret admin
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit
```

3.2 Extended ACL Configuration (SALES_POLICY on R1)

```
R1(config)# ip access-list extended PROTECT_IT
R1(config-ext-nacl)# permit icmp 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255 echo-reply
R1(config-ext-nacl)# deny ip 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)# exit

R1(config)# interface FastEthernet0/1.10
R1(config-subif)# ip access-group PROTECT_IT in
```

3.3 Switch Layer 2 Security Configurations (SW1, SW2, SW3)

```
! Port Security & STP Hardening on SW1
SW1(config)# interface e0/2
```

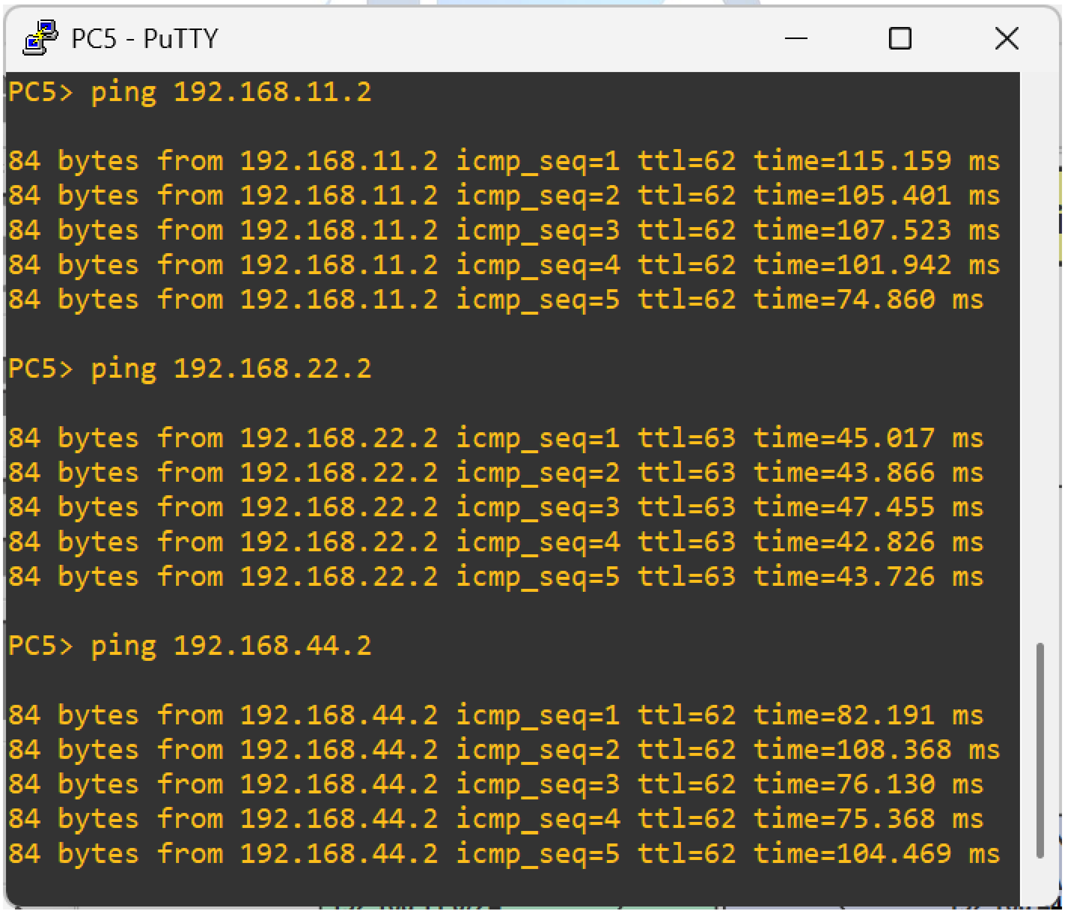
```
SW1(config-if)# switchport mode access
SW1(config-if)# switchport port-security
SW1(config-if)# switchport port-security maximum 1
SW1(config-if)# switchport port-security violation restrict
SW1(config-if)# switchport port-security mac-address sticky
SW1(config-if)# spanning-tree portfast
SW1(config-if)# spanning-tree bpduguard enable

! DHCP Snooping Setup
SW1(config)# ip dhcp snooping
SW1(config)# ip dhcp snooping vlan 10,40
SW1(config)# interface e0/1
SW1(config-if)# ip dhcp snooping trust
```

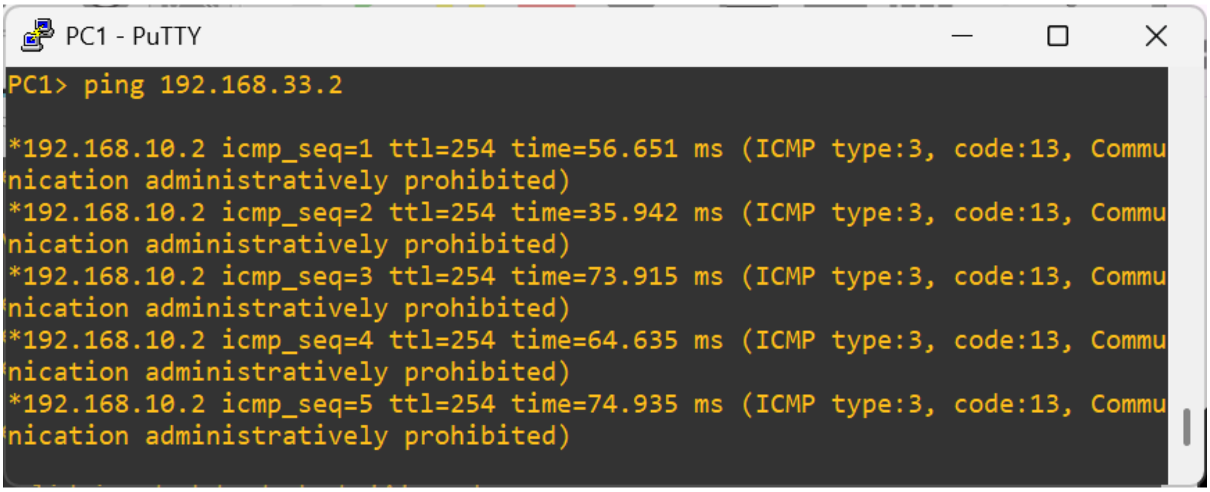
4. Verification & Testing Results

4.1 Verification 1: Access Control List Policy Enforcement

Verifies that the IT Department (VLAN 30) maintains full access to Sales (VLAN 10) and Finance (VLAN 40), while Sales and Finance are restricted from initiating connectivity toward IT.



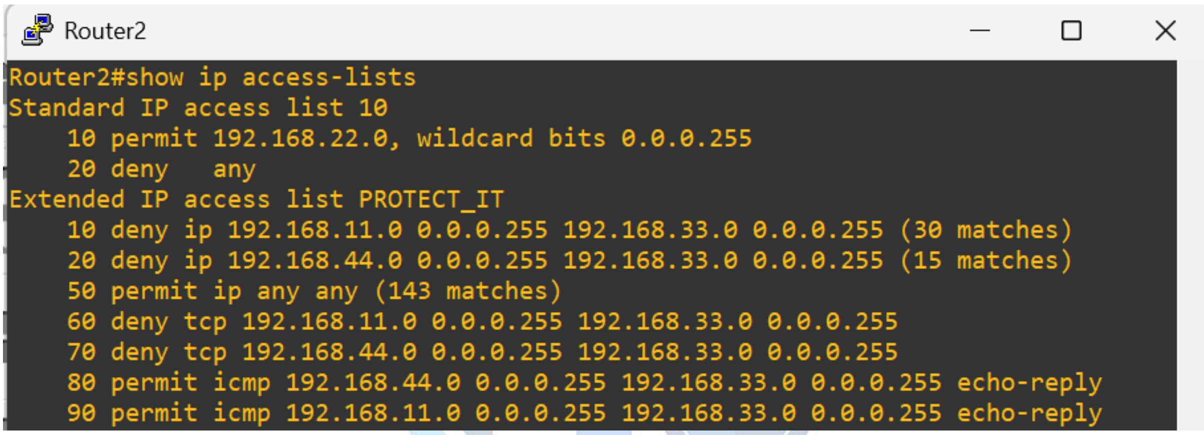
(Ping IT-PC to Sales-PC, HR-PC and Finance-PC allowed)



```
PC1> ping 192.168.33.2

*192.168.10.2 icmp_seq=1 ttl=254 time=56.651 ms (ICMP type:3, code:13, Commu
nication administratively prohibited)
*192.168.10.2 icmp_seq=2 ttl=254 time=35.942 ms (ICMP type:3, code:13, Commu
nication administratively prohibited)
*192.168.10.2 icmp_seq=3 ttl=254 time=73.915 ms (ICMP type:3, code:13, Commu
nication administratively prohibited)
*192.168.10.2 icmp_seq=4 ttl=254 time=64.635 ms (ICMP type:3, code:13, Commu
nication administratively prohibited)
*192.168.10.2 icmp_seq=5 ttl=254 time=74.935 ms (ICMP type:3, code:13, Commu
nication administratively prohibited)
```

(Denied Sales-PC to Access IT-PC)



```
Router2#show ip access-lists
Standard IP access list 10
    10 permit 192.168.22.0, wildcard bits 0.0.0.255
    20 deny   any
Extended IP access list PROTECT_IT
    10 deny ip 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255 (30 matches)
    20 deny ip 192.168.44.0 0.0.0.255 192.168.33.0 0.0.0.255 (15 matches)
    50 permit ip any any (143 matches)
    60 deny tcp 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255
    70 deny tcp 192.168.44.0 0.0.0.255 192.168.33.0 0.0.0.255
    80 permit icmp 192.168.44.0 0.0.0.255 192.168.33.0 0.0.0.255 echo-reply
    90 permit icmp 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255 echo-reply
```

(ssh Access-List of Router 2 for PROTECT_IT)

4.2 Verification 2: SSH Remote Access

Verifies encrypted CLI management session over SSH v2 from management switch SW3 to Router R2.

```
SW4# ssh -v 2 -l admin 192.168.22.250

PC74# ssh -v 2 -l admin 192.168.22.250
```

(Note:VPCS and Cisco IOS ACLs can't use this command because they are stateless)

Router2

```
Router2#show ip access-lists
Standard IP access list 10
    10 permit 192.168.22.0, wildcard bits 0.0.0.255
    20 deny   any
Extended IP access list PROTECT_IT
    10 deny ip 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255 (30 matches)
    20 deny ip 192.168.44.0 0.0.0.255 192.168.33.0 0.0.0.255 (15 matches)
    50 permit ip any any (143 matches)
    60 deny tcp 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255
    70 deny tcp 192.168.44.0 0.0.0.255 192.168.33.0 0.0.0.255
    80 permit icmp 192.168.44.0 0.0.0.255 192.168.33.0 0.0.0.255 echo-reply
    90 permit icmp 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255 echo-reply
```

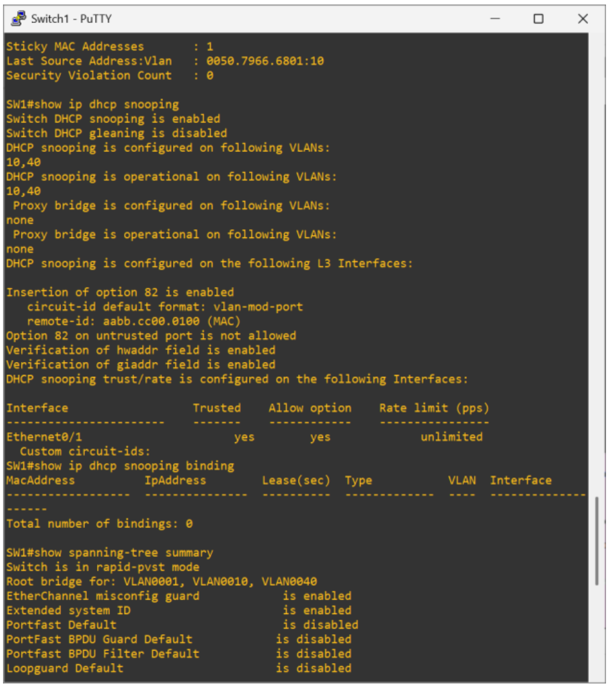
4.3 Verification 3: Layer 2 Security Features

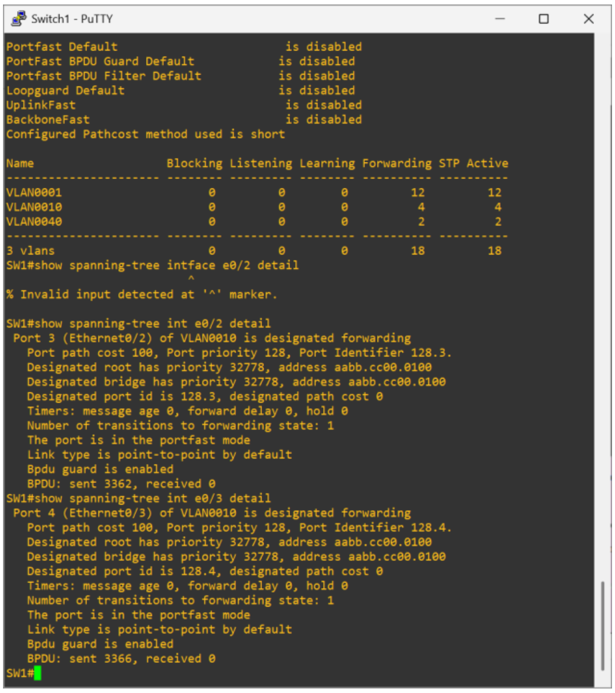
Validates Port Security bindings, DHCP Snooping database entries, and active STP Guard states across access switches.

Switch1 - PuTTY

```
SW1#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
                (Count)       (Count)          (Count)
---------------------------------------------------------------------------
      Et0/2              1            1                  0         Restrict
      Et0/3              1            1                  0         Restrict
---------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 4096
SW1#show port-security address
               Secure Mac Address Table
-------------------------------------------------------------------------------
Vlan    Mac Address       Type                          Ports    Remaining Age
                                                                    (mins)
----    -----------       ----                          -----    -------------
  10    0050.7966.6800    SecureSticky                  Et0/2         -
  10    0050.7966.6801    SecureSticky                  Et0/3         -
-------------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 4096
SW1#show port-security int e0/2
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0050.7966.6800:10
Security Violation Count   : 0

SW1#show port-security int e0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
```





(SW1: Verification of Port-Security, DHCP Snooping and Spanning-Tree)

(Note: SW2, SW3 and SW4 are included in Screenshots Folder)

5. Troubleshooting Report & Challenges Faced

5.1 Challenge 1: Stateless Extended ACL Blocking Return ICMP Traffic

- **Issue Identified:** Blocking IP/TCP traffic from Sales (VLAN 10) to IT (VLAN 30) using 'deny ip 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255' completely broke ping requests initiated by IT.
- **Technical Cause:** Cisco IOS ACLs are stateless. When IT sends an ICMP Echo Request (Type 8) to Sales, Sales receives it and generates an ICMP Echo Reply (Type 0). R1 evaluates this inbound reply from Sales against the 'deny ip' statement and drops it, causing IT's ping to time out.
- **Resolution Implemented:** Configured an explicit rule permitting 'echo-reply' packets before the blanket 'deny ip' statement:

```
ip access-list extended PROTECT_IT
 permit icmp 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255 echo-reply
 deny ip 192.168.11.0 0.0.0.255 192.168.33.0 0.0.0.255
 permit ip any any
```

5.2 Challenge 2: Absence of Native SSH Client in VPCS Nodes

- **Issue Identified:** VPCS nodes in GNS3 lack native SSH client binaries, preventing direct CLI execution of the 'ssh' command from VPCS terminals.

6. Conclusion

This assignment successfully engineered and verified a secure inter-VLAN enterprise infrastructure in GNS3. By utilizing Extended Named Access Control Lists with precise ICMP type handling, unidirectional access policies were established—allowing administrative oversight from IT while restricting unauthorized lateral movement from departmental VLANs.

Furthermore, network devices were hardened using SSH v2 with 1024-bit RSA encryption, while Layer 2 defenses (Port Security, DHCP Snooping, PortFast, and BPDU Guard) were verified to protect the physical access layer against unauthorized access points, rogue DHCP servers, and STP topology manipulation attacks.